torch.nn.functional.avg_pool2d#

https://pytorch.org/docs/stable/generated/torch.nn.functional.avg_pool2d.htm

Description:

Applies 2D average-pooling operation in kH×kWkH \times kWkH×kW regions by step size sH×sWsH \times sWsH×sW steps. The number of output features is equal to the number of input planes. See AvgPool2d for details and output shape. input – input tensor (minibatch,in_channels,iH,iW)(\text{minibatch} , \text{in_channels} , iH , iW)(minibatch,in_channels,iH,iW) kernel_size – size of the pooling region. Can be a single number, a single-element tuple or a tuple (kH, kW) stride – stride of the pooling operation. Can be a single number, a single-element tuple or a tuple (sH, sW). Default: kernel_size

Function Signature

```
torch.nn.functional.avg_pool2d(input, kernel_size,
stride=None, padding=0, ceil_mode=False,
count_include_pad=True, divisor_override=None) → Tensor#

Parameters
```

Detailed Documentation

Documentation

Applies 2D average-pooling operation in kH×kWkH \times kWkH×kW regions by step size sH×sWsH \times sWsH×sW steps. The number of output features is equal to the number of input planes.

See AvgPool2d for details and output shape.

input – input tensor (minibatch,in_channels,iH,iW)(\text{minibatch} , \text{in_channels} , iH , iW)(minibatch,in_channels,iH,iW)

kernel_size – size of the pooling region. Can be a single number, a single-element tuple or a tuple (kH, kW)

stride – stride of the pooling operation. Can be a single number, a single-element tuple or a tuple (sH, sW). Default: kernel_size

padding – implicit zero paddings on both sides of the input. Can be a single number, a single-element tuple or a tuple (padH, padW). Should be at most half of effective kernel size, that is ((kernelSize−1)∗dilation+1)/2((kernelSize - 1) * dilation + 1) / 2((kernelSize−1)∗dilation+1)/2. Default: 0

ceil_mode – when True, will use ceil instead of floor in the formula to compute the output shape. Default: False

count_include_pad – when True, will include the zero-padding in the averaging calculation. Default: True

divisor_override – if specified, it will be used as divisor, otherwise size of the pooling region will be used. Default: None